

1. Which are the advantages of using IDSs at the network level?

- a) Ensures the protection of electronic transactions, detecting unauthorized change to data
- b) Does not affect the performance of the supervised systems**
- c) Can monitor any traffic, including IPsec and SSL
- d) High scalability (with a few well-placed sensors it is possible to monitor an entire network)**

2. Which tool do you use when you need to detect the running services on a computer?

- a) Port scanner**
- b) IDS at the host level
- c) Packet sniffing
- d) File integrity checker

3. Which of the following statements about signature-based IDSs are true?

- a) Generating a relatively low number of false positive alarms**
- b) Provides detailed info about the type of attack**
- c) Requires trained personnel to investigate the alarm
- d) Can detect new types of attack

4. Which application-level protocol uses UDP for data transfer?

- a) DNS**
- b) LDAP
- c) HTTP
- d) SNMP**

5. Which of the following products can be used to scan the existing vulnerabilities on a computer system?

- a) Snort
- b) Ethereal/Wireshark
- c) Nessus**
- d) Nmap

6. What protocol is used to discover the MAC address for an IP address?

- a) ARP**
- b) ICMP
- c) DNS
- d) DHCP

7. What IP class is the IP address 190.221.78.10?

- a) Class A
- b) Class C
- c) Class D
- d) Class B**

8. Which are not routable IP addresses on the Internet?

- a) 192.169.30.20
- b) 192.128.0.10
- c) 120.15.250.5
- d) **172.31.250.24**

9. What is happening in a TCP handshake?

- a) The sender computer sends a TCP packet with SYN and ACK flag set
- b) **The sender computer sends a TCP packet with SYN flag set**
- c) **The receiver computer sends a TCP packet with ACK flag set**
- d) The receiver computer sends a TCP packet with SYN flag set

10. What is a firewall?

- a) A complete security architecture
- b) **A control and monitoring point for the network traffic**
- c) Best single solution for all network's security problems
- d) **Practical implementation of network security policy**

11. When a frame is received by a router network interface, what address is changed inside the frame?

- a) IP address
- b) **MAC address**
- c) Port address
- d) IPX address

12. DMZ is:

- a) Intranet network segment protected by a firewall, where the private services are available
- b) **A network segment between Internet and the intranet**
- c) **A network segment where public services and application gateways are deployed**
- d) A network segment in front of a firewall, which is not protected by the firewall

13. A company has a Class C and needs 5 sub-networks, each with a minimum of 20 workstations. Which subnet mask is needed?

- a) 255.255.255.240
- b) 255.255.255.0
- c) 255.255.255.92
- d) **255.255.255.224**

14. What types of firewalls maintain state information for packet analysis?

- a) **Stateful inspection firewall**
- b) Packet filter firewall
- c) Application level gateway
- d) Circuit level gateway

15. What are the differences between IPS and IDS?

- a) Operates inline
- b) May analyze encrypted traffic
- c) Detects network viruses
- d) Is implemented only in hardware

16. The network switch is better than the hub because:

- a) Provides the minimum security...
- b) It has more network ports
- c) The traffic for a certain workstation is sent to all workstations
- d) It ensures a larger bandwidth per port

17. Which fields from the IP headers are used for infinite loop prevention within the network packets?

- a) time-to-live
- b) header checksum
- c) service type
- d) fragmentation offset

2024

1. Q: The "Port Security" mechanism prevents what type of attacks?

A: MAC flooding, DHCP starvation, port stealing

2. Q: What service is on TCP port 25?

A: SMTP

3. Q: What command for UDP scanning?

A: nmap -sU

4. Q: During what type of attack are flags FIN, URG, PUSH set?

A: XMAS

5. Q: If you use arp spoof and perform a man-in-the-middle attack, if the victim logs in

on Gmail, can you see their username and password?

6. Which of the following products can be used to scan the existing vulnerabilities on a computer system

- a. Snort
- b. Ethereal-Wireshark

- c. Nessus
- d. Suricata
- e. None of the above

7. Q: Best method for protection against mitm?

A: traffic encryption

8. Q: What tools can be used for wireless attacks?

A: KRACK, Rogue Access Point

9. Q: What packets are sent when performing a three-way handshake?

A: SYN, SYN-ACK, ACK

10. Q: A packet filtering firewall operates on what OSI model layers?

A: Network, Transport

11. Q: Next-gen Firewall (NGFW) vs Stateful Inspection Firewall?

A: <https://info.support.huawei.com/info-finder/encyclopedia/en/NGFW.html>

12. Q: If an UDP Scan is performed and there is no ICMP error what can we say about the port?

A: port is open

1. The Message Integrity Code (MIC):

- a) uses the same frame format as the WEP protocol
- b) uses a frame with sequence numbers**
- c) the source and destination MAC addresses are not included in the MIC calculation
- d) uses only the frame's data for the calculation**

2. The 802.11i protocol:

- a) does not provide a robust security service
- b) uses only the AES protocol for encryption**
- c) does not support mutual authentication
- d) uses the same keys regardless of transmission time

3. Key management in the 802.11i standard is performed:

- a) only through a pairwise key
- b) during the 4-way handshake**
- c) only through a pre-shared key
- d) only through a group key

4. The CCM (CCMP) protocol:

- a) uses the RC4 algorithm
- b) uses AES**
- c) encrypts the entire CCMP frame
- d) uses a header composed only of the current packet number

5. SIM authentication:

- a) uses SIM cards for the RADIUS server
- b) does not use a shared key known by both the SIM and the GSM authentication center
- c) the authentication and encryption algorithms are stored on the RADIUS server
- d) ensures mutual authentication between the client and an AAA server**

6. The LEAP authentication protocol:

- a) performs mutual authentication
- b) uses PKI certificates
- c) is a non-proprietary protocol
- d) uses SIM modules for authentication

Answer: None

7. MAC authentication:

- a) assumes the client verifies the access point's MAC address
- b) assumes the access point verifies the client's MAC address**
- c) assumes both verifications
- d) assumes using an EAP protocol to verify MAC addresses

8. In authentication using the EAP protocol, the uncontrolled port is:

- a) a port that allows only the transmission of authentication messages**
- b) a port that allows the transmission of other data as well
- c) a port opened only after the work session ends
- d) a port that is used only for data representing MAC addresses

9. Protection against a denial-of-service (DoS) attack is achieved by:

- a) requiring end-to-end mutual authentication
- b) aiming directional antennas outward from the building**
- c) deliberately occupying the transmission channels
- d) using the victim node as an intermediary to continuously retransmit useless data packets

10. Stand-alone access points:

- a) have centralized management for transmit power and frequencies
- b) have high acquisition, installation, and operational costs
- c) can operate independently, providing authentication, security, and management in a single unit**
- d) are not recommended for small and medium businesses

11. The regeneration of encryption keys at specified intervals:

- a) ensures error-free transmission
- b) is a methodology that does not require RADIUS servers
- c) ensures simpler management of network security
- d) ensures increased data confidentiality**

12. Grouping into work domains of wireless networks:

- a) assumes creating cells only at the department level
- b) assumes a wireless service using the same SSID**
- c) assumes creating cells only with the same type of access point
- d) assumes creating cells only with the same type of client

13. The RADIUS authentication system:

- a) uses only servers with centralized databases
- b) uses only servers with distributed databases
- c) uses only servers with databases linked together
- d) uses servers with distributed or centralized databases**

14. Data link layer traffic monitoring devices can be countered by:

- a) encrypting the transmitted data**
- b) analyzing the traffic flow
- c) using powerful servers
- d) using non-overlapping channels

15. Protection against rogue (false) access point attacks:

- a) is achieved by using high operating speeds
- b) is achieved by using mutual authentication**
- c) is achieved by using multi-system access points
- d) is achieved by using cells by domain

16. In the case of wireless computer networks, we can have:

- a) passive attacks
- b) active attacks
- c) passive and active attacks**
- d) topological attacks

17. A session hijacking attack involves:

- a) altering packets from the access point to the client
- b) copying the access point's MAC address, de-authenticating the client, and relaying data between the attacker and tc) disabling the access point's operation**
- d) establishing the link between the attacker and the client

18. A man-in-the-middle attack involves:

- a) the attacker becoming an access point for the client and a client for the real access point**
- b) the attacker changing the SSID of the access point
- c) creating a new cell
- d) extending the existing cell

19. The WEP protocol:

- a) uses a different algorithm for the sent message and the received message
- b) does not have a field to protect against unauthorized modifications
- c) encrypts all fields of the frame
- d) does not encrypt the IV field**

20. The SSID identifier:

- a) is an identifier only for the access point
- b) is an identifier only for the client
- c) is an encrypted message in the Beacon frames
- d) is an identifier that provides the ability to logically separate wireless networks**

21. Shared-key authentication:

- a) presumes authentication of the client**
- b) presumes authentication of the access point
- c) presumes authentication of both the access point and the client
- d) presumes that the access point has an authentication database

22. The Initialization Vector (IV) in the WEP protocol:

- a) this field is used to calculate the ICV field
- b) it is a field that is not added to the static key, being used autonomously
- c) it is an optional field
- d) it is a field that is added to the static key**

23. The 802.1X authentication standard:

- a) is a port-based standard with an authentication server**
- b) is a standard based only on ports
- c) is a standard based only on the authentication server
- d) is a standard based on device authentication

24. The EAP protocol uses:

- a) only lower-level EAPOL packets
- b) only application-level RADIUS packets
- c) both types of packets**
- d) is used only by wireless networks

25. The EAP-MD5 protocol:

- a) is a certificate-authority-based protocol
- b) protects the message exchange by creating a unique fingerprint that is digitally signed
- c) uses PKI certification
- d) allows elimination of man-in-the-middle attacks

Answer: None

26. The Extended Authentication Protocol with Transport Layer Security (EAP-TLS):

- a) is a protocol based on PKI certificates, requiring both the server and the client to present certificates**
- b) does not perform mutual authentication
- c) does not rely on X.509 certificates
- d) is based on username and password authentication

27. The Protected Extensible Authentication Protocol (PEAP):

- a) ensures authentication based on username and password**
- b) does not support mutual authentication
- c) the PEAP client authenticates directly to the authentication server
- d) does not support fragmentation

28. The WPA protocol:

- a) uses WEP-based encryption
- b) provides confidentiality using the TKIP protocol**
- c) ensures message integrity using the ICV
- d) uses an initialization vector like WEP

29. The Temporal Key Integrity Protocol (TKIP):

- a) uses single-phase key mixing
- b) does not use a sequence counter

c) uses the MIC for message integrity

d) encrypts the entire MPDU frame, including the header